

CE 310

Week 12 — Sensitivity Analysis

How Uncertainty in Inputs Propagates to Outputs

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Why Sensitivity Analysis?

You now have a regression model that predicts cooling energy from outdoor temperature and occupancy. To use it for engineering design, you need to know: **if an input is wrong by some amount, how wrong is the prediction?**

Inputs are always uncertain in practice:

- Outdoor temperature forecast for next summer: $\pm 5^{\circ}\text{F}$ from reality is typical
- Occupied hours per year: schedule changes by dozens of hours from year to year
- The regression coefficients themselves carry uncertainty (standard errors)

Sensitivity analysis answers: *if one input changes by X , how much does the output change?*

Why Sensitivity Analysis? — The Engineering Decision

This week: three complementary methods — analytical, one-at-a-time perturbation, and tornado diagrams. Next week (Monte Carlo) will extend this to full probability distributions.

For a Linear Model, the Slope IS the Sensitivity

For any linear model $\hat{y} = \beta_0 + \beta_1 T + \beta_2 \cdot BH$:

Sensitivity to temperature = β_1

This is just the definition of a slope: $\frac{\partial \hat{y}}{\partial T} = \beta_1$. No matter what temperature you start from, each 1°F increase raises $\hat{y}$ by exactly β_1 kWh/hr.

Why this matters:

- Sensitivity is constant across the entire temperature range — no calculus needed
- The slope you already computed in Week 10 is the sensitivity
- Sensitivity to BH is β_2 — the occupancy premium you already know

Slope as Sensitivity — Applied to Cooling

This is the cleanest case: linear models have constant sensitivity. The complexity comes when you have interaction terms (sensitivity then depends on another variable) or non-linear relationships.

One-at-a-Time (OAT) Perturbation

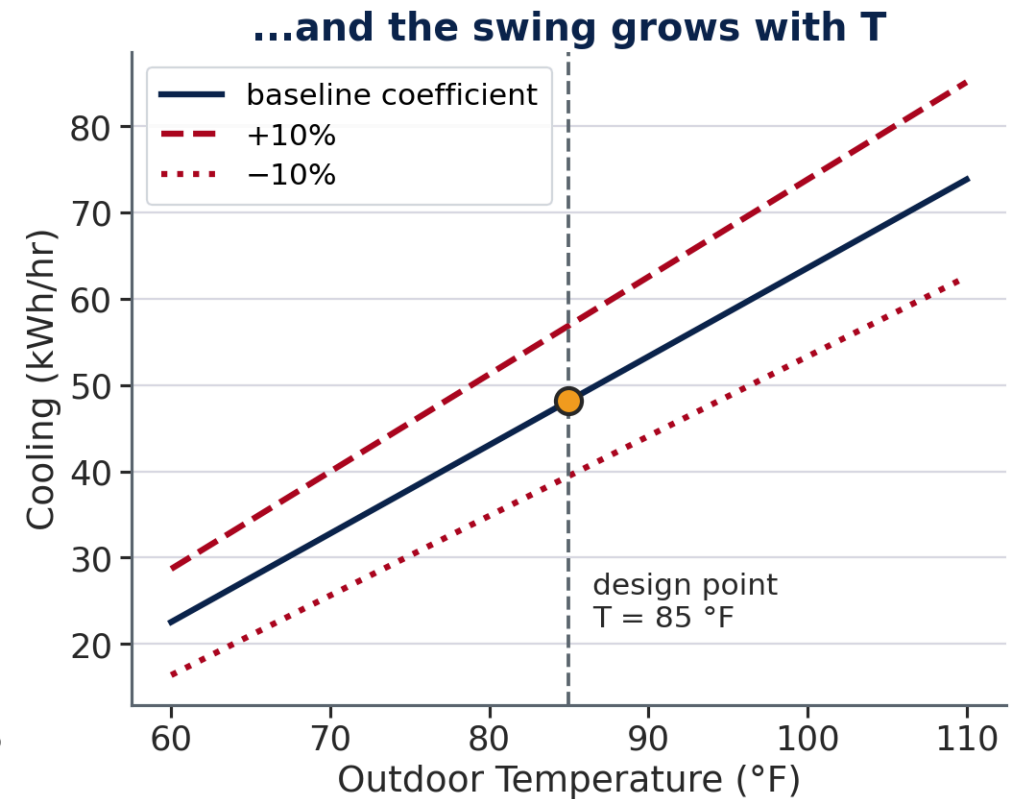
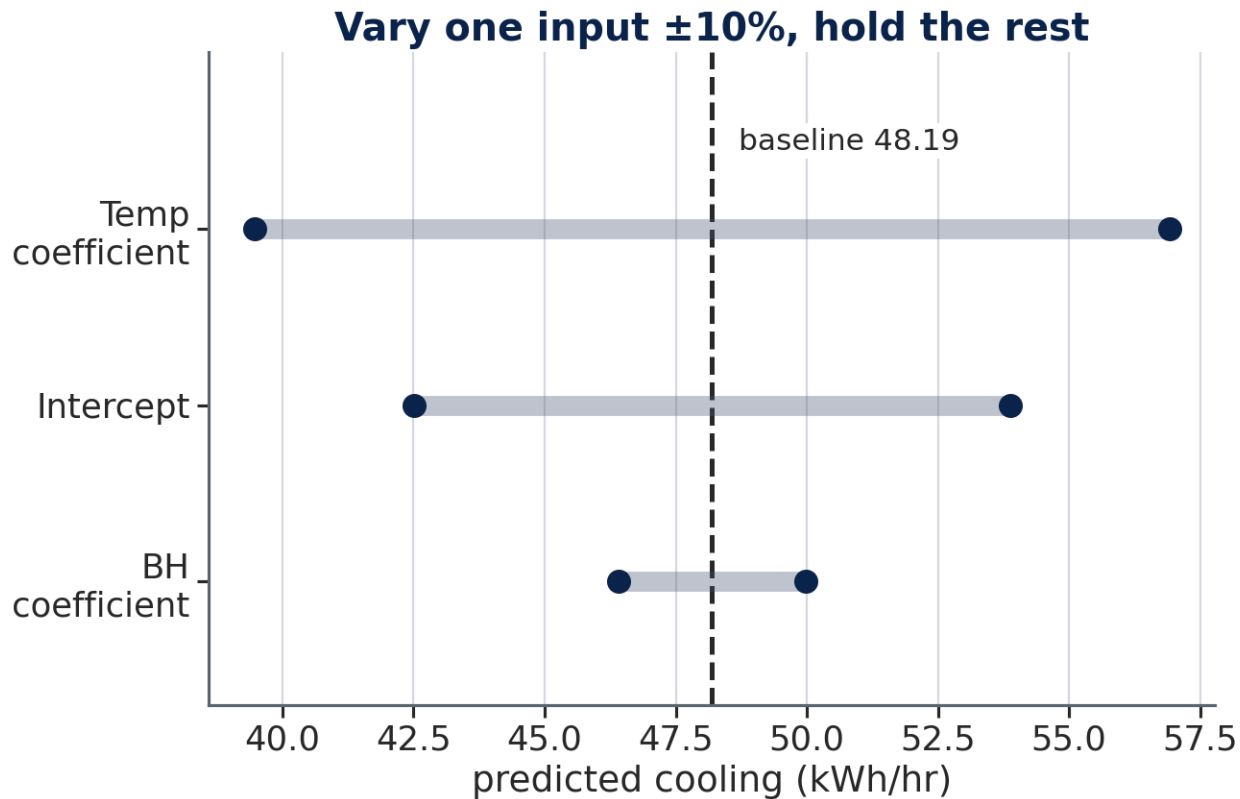
OAT sensitivity tests: **change one input by a small amount, hold everything else fixed, measure the output change.**

Example at $T = 85^{\circ}\text{F}$, $\text{BH} = 1$ (baseline ≈ 49.6 kWh/hr):

- Increase temperature by $+5^{\circ}\text{F} \rightarrow$ prediction at $T = 90^{\circ}\text{F}$
- $\text{Change} = 5 \times \beta_1 = 5 \times 1.03 = \mathbf{5.13 \text{ kWh/hr}}$ (+10.3%)

OAT gives a direct, interpretable message: a 5°F forecast error changes the peak cooling prediction by about 5 kWh/hr. Engineers can judge whether that is acceptable given the chiller they are sizing.

What OAT Actually Does



The left panel is the whole method: nudge one coefficient, hold the rest, record the swing. The right panel is the catch — the same $\pm 10\%$ moves the answer more at 105 °F than at 65 °F.

Annual Aggregate Sensitivity

A single prediction changes by $\beta_1 \times \Delta T$ for a 1-hour shift. But an annual energy budget involves 8,760 predictions — one per hour.

If outdoor temperatures are systematically biased by **+5°F** (e.g., the weather station reads 5°F warmer than the true value):

$$\text{Total annual error} = n \times \Delta T \times \beta_1 = 8,760 \times 5 \times 1.03 \approx 45,100 \text{ kWh/yr}$$

To put this in context: the annual cooling total is roughly 166,000 kWh. A 5°F systematic bias produces a ~27% overestimate of annual cooling energy.

This is why typical meteorological year (TMY) data — which represents an average year — can significantly underpredict or overpredict cooling energy in a year that is warmer or cooler than average. Sensitivity analysis quantifies that risk.

Quick Check — Which Parameter Dominates?

90 seconds · rank A, B, C

Model 2 has three parameters. At a design point of $T = 85^{\circ}\text{F}$, $\text{BH} = 1$ (occupied):

Parameter	Value	Your $\pm 10\%$ swing
A Temperature slope	1.026 kWh/ $^{\circ}\text{F}$	± 0.103 kWh/ $^{\circ}\text{F}$
B Intercept	-56.87 kWh	± 5.69 kWh
C BH coefficient	17.9 kWh/hr	± 1.79 kWh/hr

Which parameter's 10% uncertainty produces the largest swing in the predicted cooling demand?

✓ Temperature Slope Dominates

Impact = (parameter uncertainty) × (what it multiplies)

Parameter	±10% uncertainty	× input	Swing in prediction
A Temperature slope	±0.103 kWh/°F	× 85°F	±8.75 kWh/hr ← largest
B Intercept	±5.69 kWh	× 1	±5.69 kWh/hr
C BH coefficient	±1.79 kWh/hr	× 1 (occupied)	±1.79 kWh/hr

The principle: sensitivity = uncertainty × leverage. Higher leverage amplifies even small uncertainties into large output swings.

💡 This is why the tornado diagram sorts by swing magnitude, not by parameter magnitude or uncertainty magnitude alone.

Tornado Diagram — Ranking Parameter Uncertainty

Perturb each parameter by $\pm 10\%$, compute the output swing ($|\text{pred_hi} - \text{pred_lo}|$), sort.

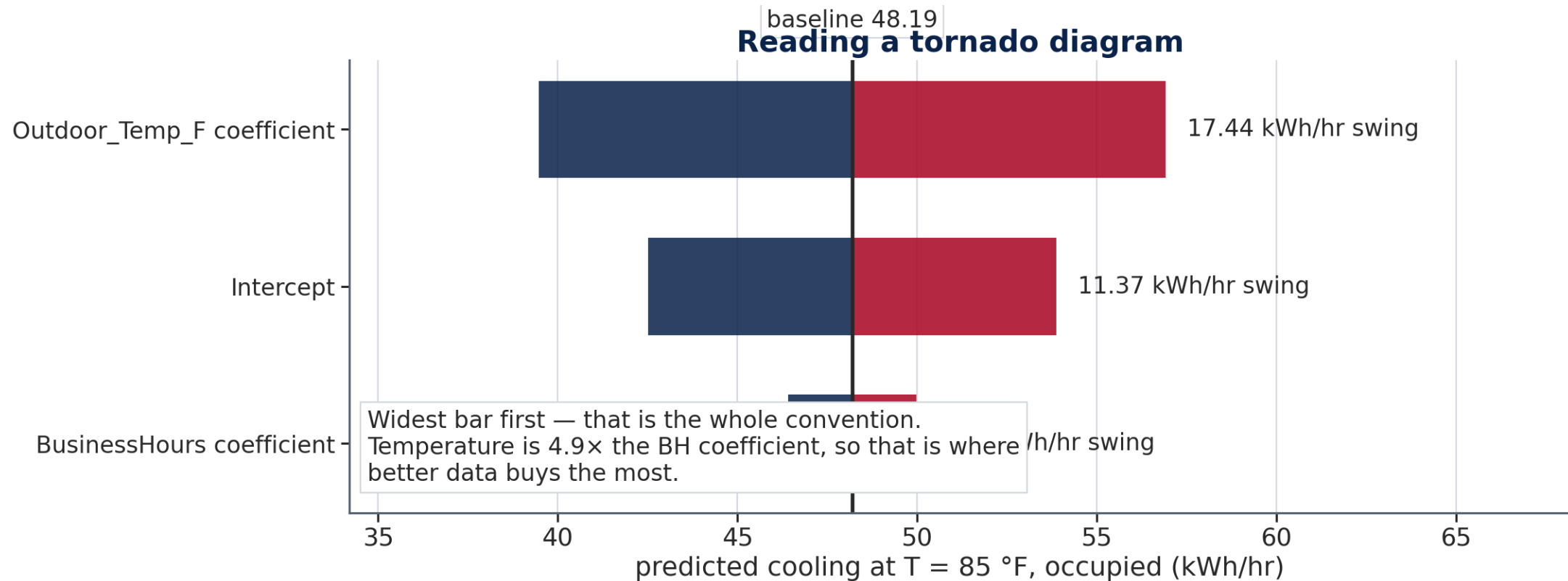
Design point: $T = 85^\circ\text{F}$, $\text{BH} = 1$ (Model 2)

Parameter	Swing (kWh/hr)
Outdoor_Temp_F slope	17.5 kWh
Intercept	11.4 kWh
BH coefficient	3.6 kWh

The slope dominates because it multiplies $T = 85^\circ\text{F}$ — a large number. BH's swing is small relative to the other two.

Engineering implication: invest measurement effort in outdoor temperature accuracy before worrying about occupancy scheduling precision.

Reading a Tornado Diagram



Bars sorted widest first, all measured at the same operating point. The name comes from the shape, and the shape is the point.

Interaction Model: Sensitivity Depends on Occupancy

The additive Model 2 has one slope for all hours. The interaction Model 5 has two:

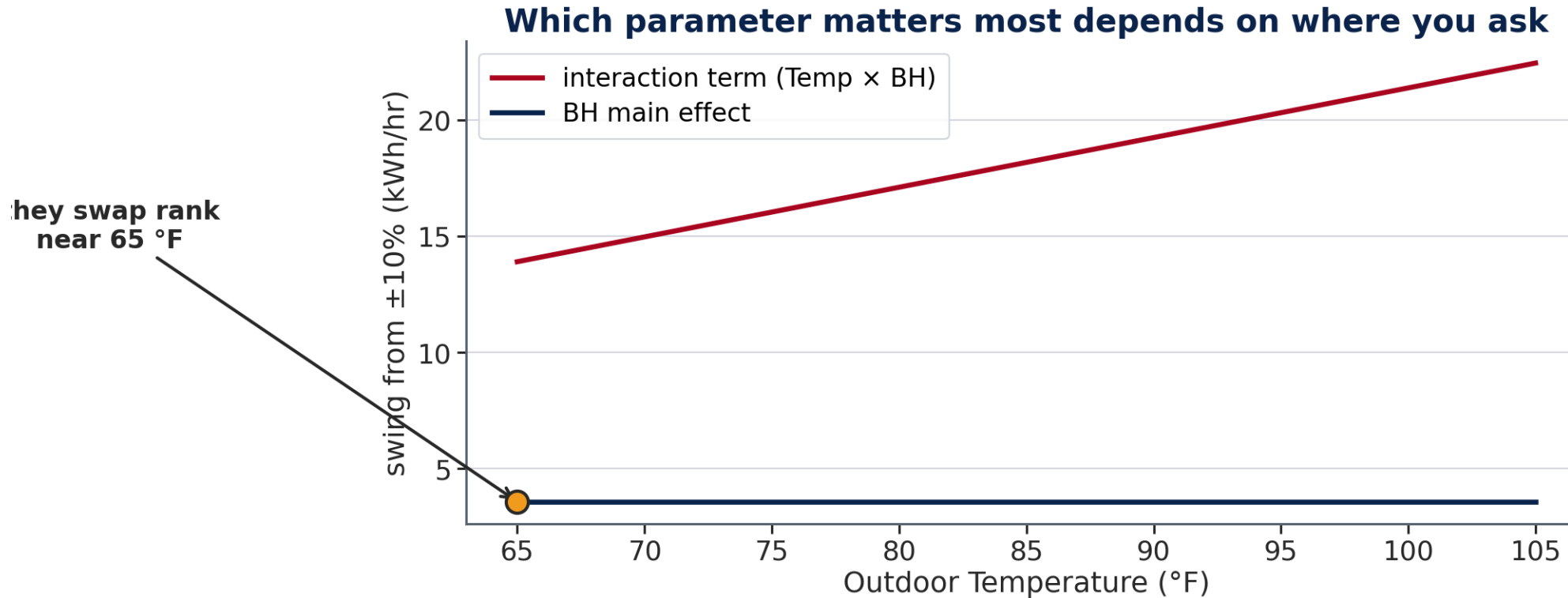
Condition	Temperature Sensitivity
Unoccupied (BH = 0)	0.71 kWh/°F
Occupied (BH = 1)	1.78 kWh/°F

The occupied building is **2.5× more sensitive** to outdoor temperature.

For a +5°F forecast error:

- Unoccupied: $0.71 \times 5 = \mathbf{3.6 \text{ kWh/hr}}$ shift
- Occupied: $1.78 \times 5 = \mathbf{8.9 \text{ kWh/hr}}$ shift

The Ranking Is Not Fixed



In the interaction model, the temperature-interaction term's swing grows with temperature while the occupancy main effect's does not. Below the crossing point one dominates; above it, the other.

Sensor Accuracy — How Good Is Good Enough?

Sensitivity analysis gives a practical sensor specification: how accurate does a temperature sensor need to keep cooling predictions within a stated tolerance?

At $T = 85^{\circ}\text{F}$, $BH = 1$ (prediction ≈ 49.6 kWh/hr):

Tolerance	Allowed error (kWh)	Required sensor accuracy
10%	4.96 kWh/hr	$\pm 4.8^{\circ}\text{F}$
1%	0.50 kWh/hr	$\pm 0.48^{\circ}\text{F}$

Standard weather station sensors ($\pm 1^{\circ}\text{F}$) easily meet 10% tolerance. For 1% tolerance, they fall short — justifying investment in higher-accuracy instrumentation for precise energy benchmarking.

Quick Check — Accuracy Spec

Work it out · 60 seconds

The annual energy model has baseline $\approx 166,000$ kWh/yr.

Each 1°F of temperature forecast bias shifts annual cooling energy by about **9,020 kWh/yr**.

If you want to keep annual forecast error under 10%, what temperature accuracy do you need?

- **A) $\pm 1.8^\circ\text{F}$**
- **B) $\pm 5^\circ\text{F}$**
- **C) $\pm 10^\circ\text{F}$**

✓ Answer: A — $\pm 1.8^{\circ}\text{F}$ Maximum Bias

Sensitivity analysis converts to an engineering specification

Target	Calculation	Value
10% error tolerance	$0.10 \times 166,000 \text{ kWh/yr}$	16,600 kWh/yr max error
Error per 1°F of bias	$45,100 \text{ kWh} / 5^{\circ}\text{F}$	9,020 kWh/$^{\circ}\text{F}$
Maximum allowable bias	$16,600 / 9,020$	$\approx 1.8^{\circ}\text{F}$

Sensor spec implication: Professional HVAC sensors ($\pm 0.5^{\circ}\text{F}$) pass; low-cost IoT sensors (± 2 – 3°F) do not.

💡 Sensitivity analysis doesn't just tell you which input matters — it tells you *how accurately* you need to measure it. This converts a mathematical result directly into a procurement decision.

Limitations of OAT Analysis

OAT sensitivity has important limitations that motivate Monte Carlo (Week 13):

- **Only tests one input at a time:** OAT cannot capture simultaneous uncertainty across multiple inputs.
- **Single-scenario thinking:** OAT reports one perturbation result, not a probability distribution or confidence interval.
- **Linear extrapolation only:** exact for linear models, but an approximation that may fail for large perturbations in non-linear models.

OAT Limitations — What This Means in Practice

Monte Carlo simulation resolves all three: it draws thousands of random input combinations from probability distributions, runs the model for each, and builds a full distribution of outputs. The result is a **prediction interval** rather than a single "what if" number.

Bridge to Machine Learning

Sensitivity concept	ML equivalent	Connection
Slope = analytical sensitivity	Feature importance (linear models)	Coefficient $\times$ input range
OAT perturbation	Partial dependence plot (PDP)	Change one feature, hold others
Tornado diagram	Feature importance ranking	Which feature drives output most?
Interaction sensitivity	SHAP interaction values	Occupied vs. unoccupied slope gap

In ML, **feature importance** is precisely the question sensitivity analysis asks: which input variable drives the output most? SHAP (SHapley Additive exPlanations) is the modern equivalent of a tornado diagram — it decomposes each prediction into contributions from each feature.

Looking Ahead — Week 13: Monte Carlo Simulation

Sensitivity analysis answered: *if one input shifts by X , output shifts by Y .*

Monte Carlo answers: *if inputs have probability distributions, what is the full distribution of outputs?*

The method: instead of testing $\pm 5^\circ\text{F}$ as one scenario, draw 10,000 random temperature values from a realistic distribution (e.g., $\text{Normal}(\mu, \sigma)$), run the model for each, and histogram the results.

The output is a **prediction interval** — for example: "there is a 95% probability that peak cooling demand will fall between 55 and 82 kWh/hr." This is the information an engineer needs to choose a chiller capacity that balances over-sizing cost against reliability risk.